

Straight Leg Raises

Passive ROM: Therapist holds the ankle and flexes the hip joint keeping the knee in flexion.

Active ROM: The therapist instructs the patient to bend the hip keeping the knee flexed and foot raised off the plinth. The patient is guided to take the knee towards the chest as much as possible.

Performance:

- Patient lies flat on their back
- One leg fully extended (the leg being raised)
- Other leg bent at the knee, foot flat on the bed (for back support)
- Patient locks the knee of the extended leg by contracting the quadriceps
- Raises the straight leg upward to a comfortable height
- Holds briefly
- Lowers back down with control

Camera Setup:

Position → side view, perpendicular to patient

Side → camera on the side of the MOVING leg (the leg being raised) the resting bent leg sits further from camera, behind the moving leg

Height → hip height

Distance → 2 to 2.5 metres

Background → plain wall, no clutter

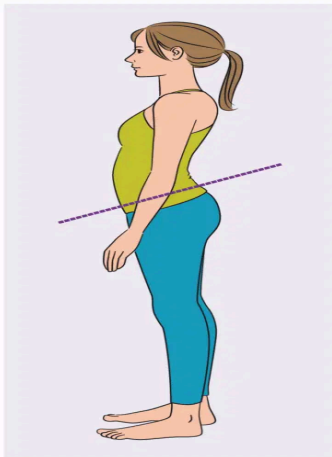
Lighting → even, no harsh backlight or window glare behind patient

Clothing → fitted dark trousers, bare feet

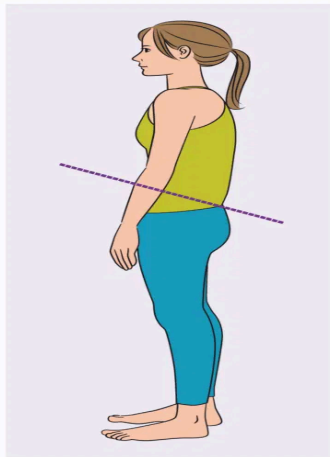
PELVIC TILT

ANTERIOR VS. POSTERIOR

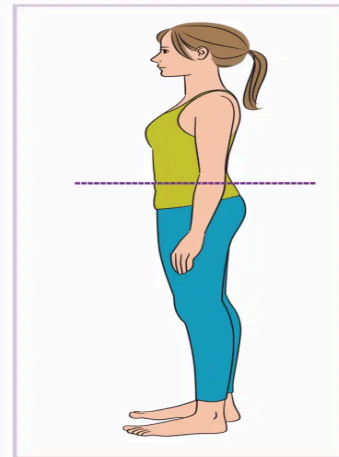
Pelvic tilt is a common condition that can cause bad posture, leading to problems with your spinal discs and increasing your risk of injury, especially when exercising.



ANTERIOR TILT
The pelvis is tilted forward.

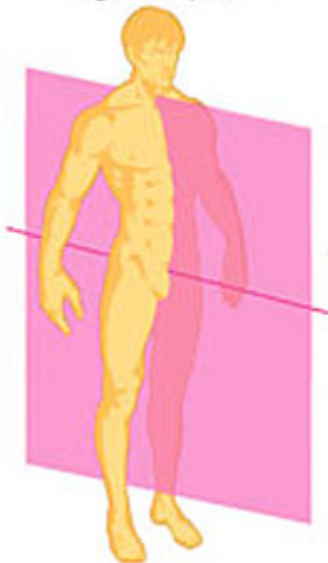


POSTERIOR TILT
The pelvis is tilted backwards.

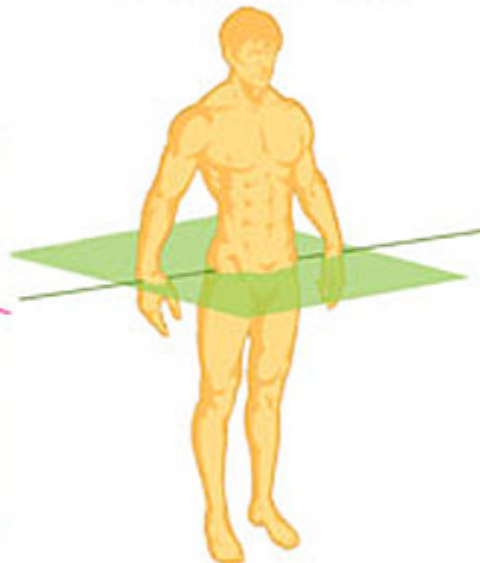


GOOD POSTURE

Sagittal plane



Horizontal plane



Frontal plane



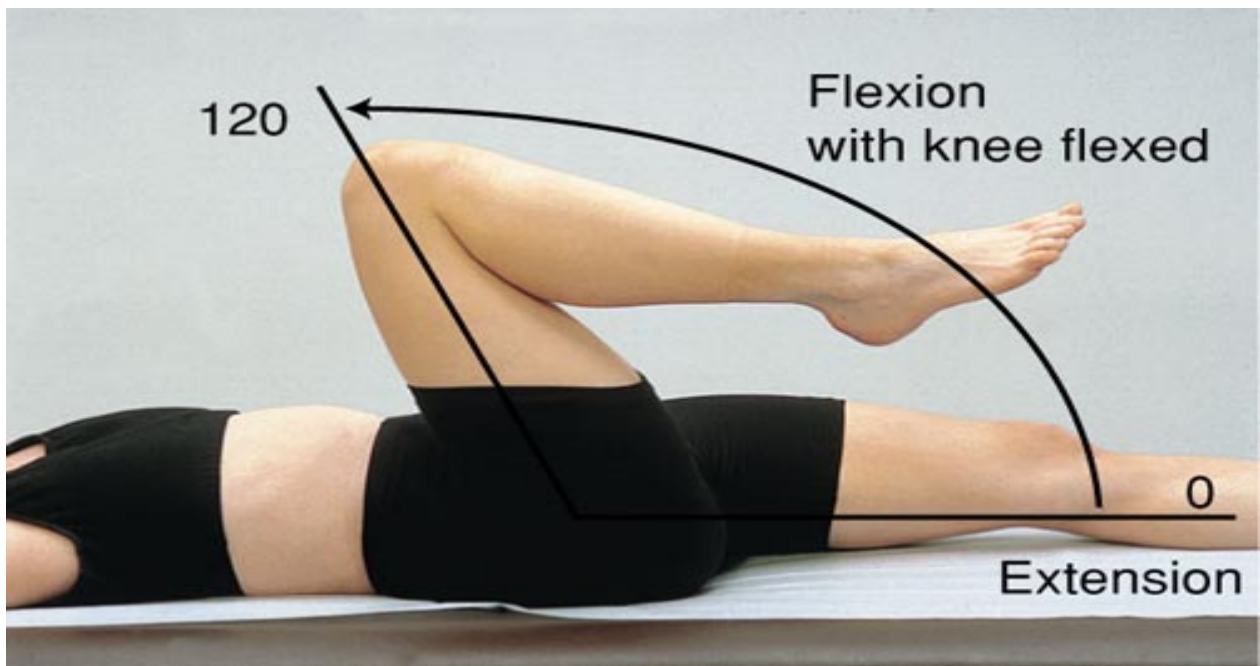
Key Angles:

1) Hip Flexion Angle (Primary Tracking Angle)

- The hip flexion angle measures the degree to which the thigh is brought closer to the torso in the **sagittal plane**. A normal range of motion is typically between

110° and 120° when the knee is bent, and can reach up to 140° depending on the individual's baseline flexibility.

- Motions of the pelvis on the lumbar spine during measurement of hip flexion or extension can artificially inflate the range of motion measurement obtained.
- Hip flexion range of motion is dependent on the position of the **knee** during movement.
- Full hip flexion is obtained only with the knee flexed.
- If the hip is flexed with the knee extended, tension in the hamstring muscles limits the motion.



2) Pelvic gap — compensation check

It is necessary for the back to be firmly pressed against the floor.

Lower back arching or lifting off the surface is not suitable and is called the **anterior pelvic tilt** or lumbar hypertension. This is one of the most common compensation errors which should be avoided.

Another major form of compensation is the **pelvic hiking sideways**. This means that one side of the hip hitches upward toward the armpit, making the waist uneven. Instead of the pelvis staying perfectly flat & level, the hip of the moving leg lifts up.

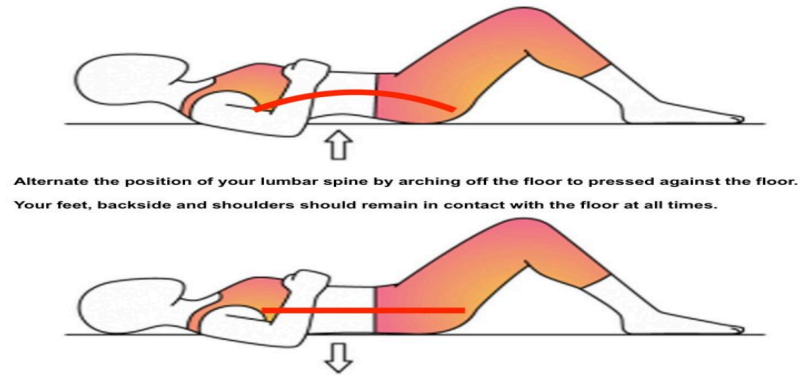
Why does this error occur?

Ans: When the knee joint is stiff & refuses to bend, the body looks for an easier way to pull the foot backward. By hiking the hip, the patient artificially shortens

the length of his/ her leg. This allows the foot to move closer to the body without forcing the knee joint itself to actually bend any further.

This error tricks the patient into thinking his/ her knee flexibility is improving when it is actually staying exactly the same. It overworks a lower back muscle called the quadratus lumborum which can quickly lead to lower back pain or muscle spasms.

- **Anterior Pelvic Tilt happens** because your **core and glutes are weak**, or your hip flexors are too tight. Your body defaults to arching your lower back to cheat its way through an exercise.
- **Pelvic Hiking happens** because your **knee or hip lacks the flexibility to bend or clear the floor**. Your body hitches the hip upward to artificially shorten the leg so you can pull or swing it forward.



3) Knee angle on moving leg — critical secondary angle

- Landmarks → hip (24) - knee (26) - ankle (28)
- Should stay → $\sim 175\text{-}180^\circ$ throughout the entire movement
- This is your single most important correctness feature
- In a Straight Leg Raise (SLR), the **knee angle of the moving leg** must remain **perfectly straight (fully extended at $\sim 180^\circ$)**. Any bend at the knee compromises the targeted hip flexor and quadriceps activation. The active leg is typically lifted to an angle of about 45° to 60° off the floor.

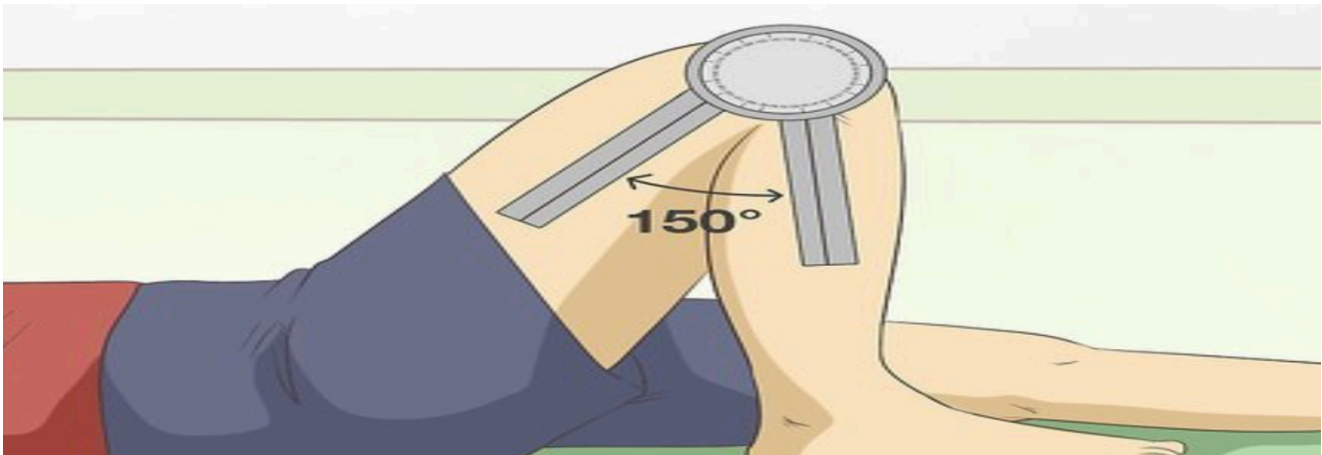
4) Ankle angle — secondary

- Landmarks → knee (26) - ankle (28) - foot (32)
- Detects foot position changes — pointing/flexing the foot
- to generate momentum is a subtle compensation pattern
- Ankle Dorsiflexion (Toes pulled toward your shin): This angle creates maximum passive tension through the posterior chain. It actively pulls and stretches the

sciatic nerve and hamstrings, which is excellent for assessing nerve root irritation, sciatic tension, or hamstring tightness.

5) Resting leg knee angle — secondary

- Landmarks on the OTHER leg → hip - knee - ankle of resting leg
- Should stay roughly constant (already bent, should not change)
- Detects resting leg sliding or shifting during the exercise



States:

S1 → leg flat, hip angle near maximum (~180°)

S2 → leg ascending, hip angle decreasing

S3 → peak height, hip angle at minimum for this rep

S4 → leg descending, hip angle increasing

Errors in Motion:

Error 1 — Knee bending during the raise

Patient cannot maintain quad contraction throughout range

Knee flexes instead of staying locked

Feature signature → high knee_std in S2/S3/S4, knee_min drops well below 175°

Error 2 — Hip hiking / pelvic compensation

Patient tilts pelvis upward on the moving side to assist the lift
instead of using true hip flexor strength

Feature signature → pelvic_gap changes significantly from S1 baseline

Error 3 — Trunk rotation / rolling

Patient rotates the trunk toward the resting leg side to "cheat" the lift

Feature signature → shoulder landmark y-position shifts unexpectedly,

or hip-shoulder relationship changes from S1 baseline

Error 4 — Resting leg sliding

Patient's bent resting leg slides or straightens during the effort

Feature signature → resting leg knee_angle changes from its starting value

Error 5 — Uncontrolled/jerky movement

Patient throws the leg up using momentum rather than controlled muscular effort, or drops it on the way down

Feature signature → high hip_std in S2 and/or S4

Error 6 — Foot not held neutral (using momentum via ankle)

Patient flexes/points the foot sharply to generate swing momentum

Feature signature → high ankle_std during S2

Error 7 — Correct (no errors)

Clean form regardless of height reached

Data Collection Protocol:

Per volunteer, per session, record:

5 reps → correct, varying heights

5 reps → knee bending error

5 reps → hip hiking error

5 reps → trunk rotation error

5 reps → resting leg sliding error

5 reps → jerky/uncontrolled error

5 reps → foot/ankle momentum error

Total → 35 reps per volunteer per session